

JudgeGuard

A correct final answer can hide a broken agent trace.

96 / 100

correct numeric-pair rankings

4 ties

retained as failures

\$0 paid API

existing local GPU; electricity
extra

Problem and business decision

An AI system can return the expected answer while recording an inconsistent tool execution. A judge can also give a corrupted answer the same score as a correct one. JudgeGuard makes those failure modes inspectable before a team trusts an evaluation score.

Data and implementation

The workbench replays four bounded, pure tools: lookup, calculator, date difference and unit conversion. It flags mismatched arguments/results without running arbitrary code. A separate pilot pairs 100 source-supported numeric answers with constructed corruptions from 28 SQuAD v2 articles. Raw judgments, source spans, hashes and pinned model revisions preserve provenance.

Measured result

Qwen3-4B Q4_K_M produced 200 real judge responses on an existing RTX 4060 Laptop GPU. It ranked 96 reference answers above their corrupted counterparts, with four ties and no parsing failures. The 95% article-cluster bootstrap interval is 92.45%-99.03%. All failures remain visible; this is not general RAG accuracy.

What a visitor can use

Edit a trace and locate the inconsistent step, then inspect the saved benchmark and its raw evidence without an API key. CLI and API expose the same audit workflow. Optional live Qwen3-0.6B inference uses ZeroGPU quota and is labeled as an uncalibrated opinion from a different model.

Business value, cost and limits

The demonstrated value is trace-level debugging and reproducible review of evaluator failures, not measured production savings. The frozen pilot used no paid API calls; existing hardware and electricity still have costs. Constructed numeric errors are narrow, public-benchmark contamination is possible, and independent human label review remains pending. Calls replay independently, so passing does not prove a complete or appropriate agent plan.

Workbench: huggingface.co/spaces/RugFace/judgeguard-app

Code and raw evidence: github.com/RyanSingh0/judgeguard (results/pilot_live.json)

Source: SQuAD v2, Rajpurkar, Jia, Liang et al.; Wikipedia passages, CC BY-SA 4.0. Pilot recorded September 17, 2026.